Through the late 1940s, the 1950s, and into the 1960s, the three teams of experimental physicists (Jodrell Bank, Cambridge, and Australia) competed with each other in building ever larger and more sophisticated radio interferometers, with ever improving resolutions. The first crucial benchmark, the 100-fold improvement necessary to begin to stir an interest among optical astronomers, came in 1949, when John Bolton, Gordon Stanley, and Bruce Slee of the Australian team produced 10-arc-minute-sized *error boxes* for the positions of a number of radio sources; that is, when they identified 10-arc-minutesized regions on the sky in which the radio sources must lie. (Ten arc minutes is one-third the diameter of the Sun as seen from Earth, and thus much poorer resolution than the human eye can achieve with light, but it is a remarkably good resolution when working with radio waves.) When the error boxes were examined with optical telescopes, some, including Cyg A, showed nothing bright of special note; finer radio resolution would be needed to reveal which of the plethora of optically dim objects in these error boxes might be the true sources of the radio waves. In three of the error boxes, however, there was an unusually bright optical object: one remnant of an ancient supernova, and two distant galaxies.

As difficult as it may have been for astrophysicists to explain the radio waves that Jansky had discovered emanating from our own galaxy, it was even more difficult to understand how distant galaxies could emit such strong radio signals. That some of the brightest radio sources in the sky might be objects so extremely distant was too incredible for belief (though it ultimately would turn out to be true). Therefore, it seemed a good bet (but those who made the bet would lose) that each error box's radio signals were coming not from the distant galaxy, but rather from one of the plethora of optically dim but nearby stars in the error box. Only better resolution could tell for sure. The experimental physicists pushed forward, and a few optical astronomers began to watch with half an eye, mildly interested.

By summer 1951, Ryle's team at Cambridge had achieved a further 10-fold improvement of resolution, and Graham Smith, a graduate student of Ryle's, used it to produce a i-are-minute error box for Cyg A—a box small enough that it could contain only a hundred or so optical objects (objects seen with light). Smith airmailed his best-guess position and its error box to the famous optical astronomer Walter Baade at the Carnegie Institution in